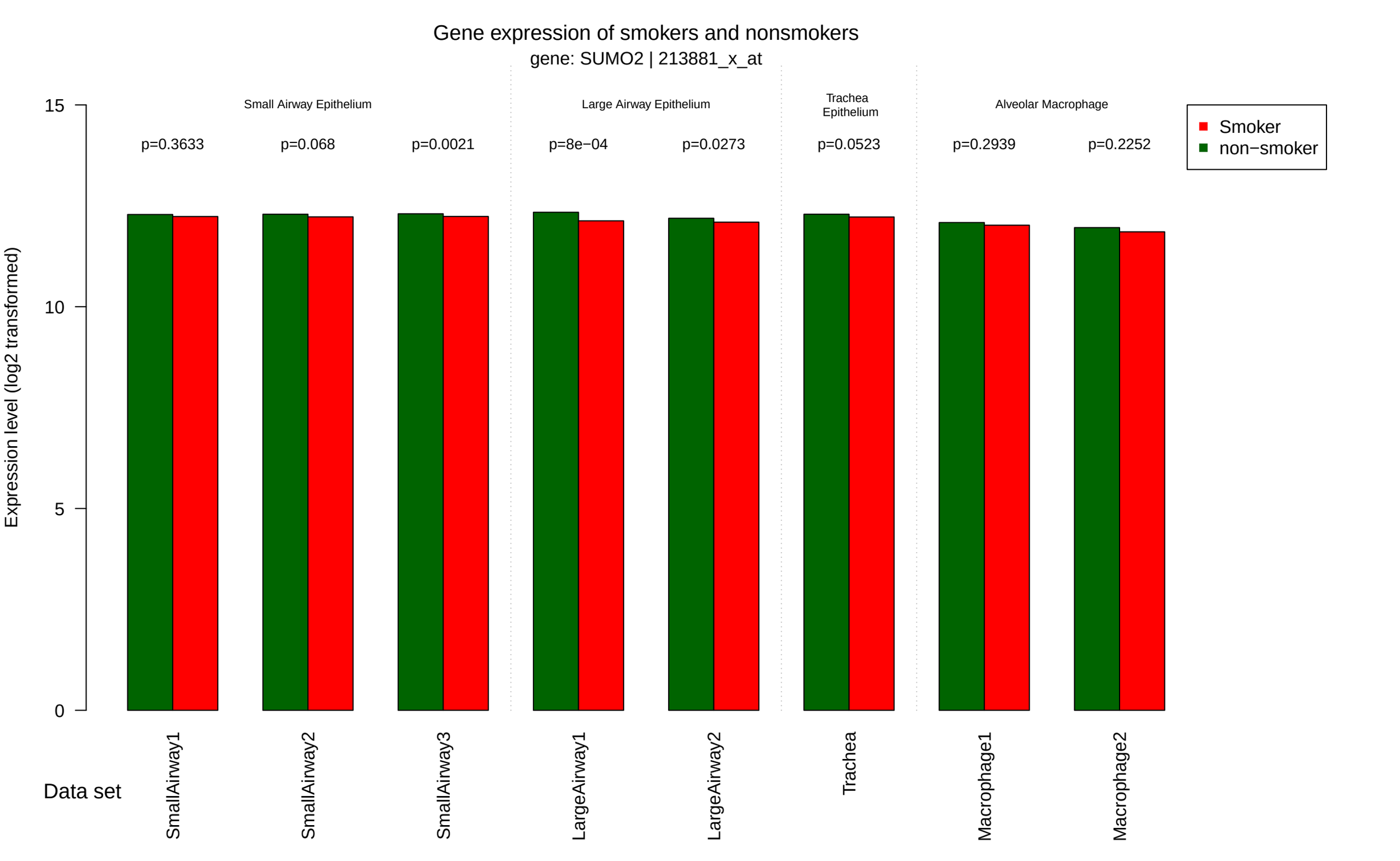
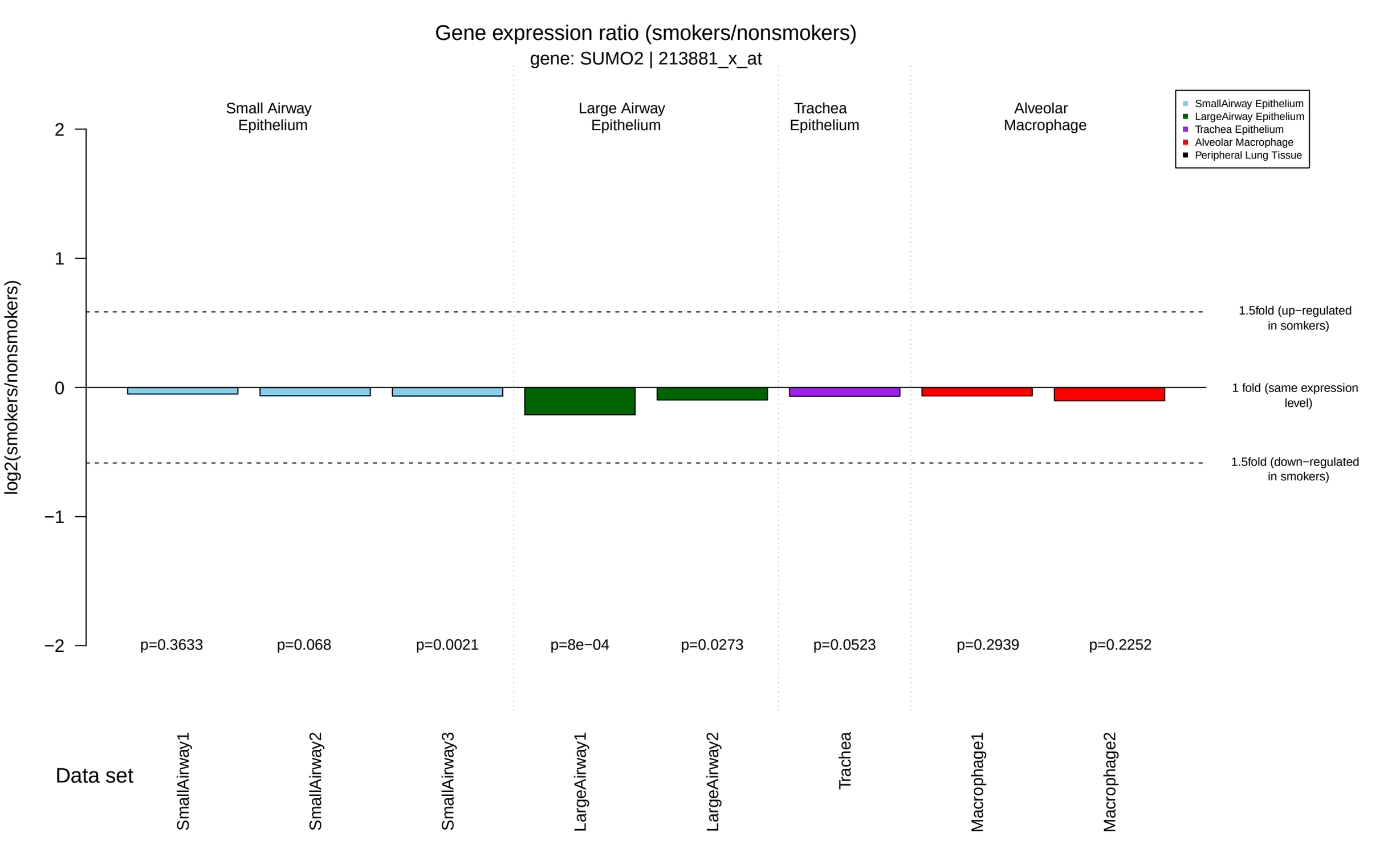
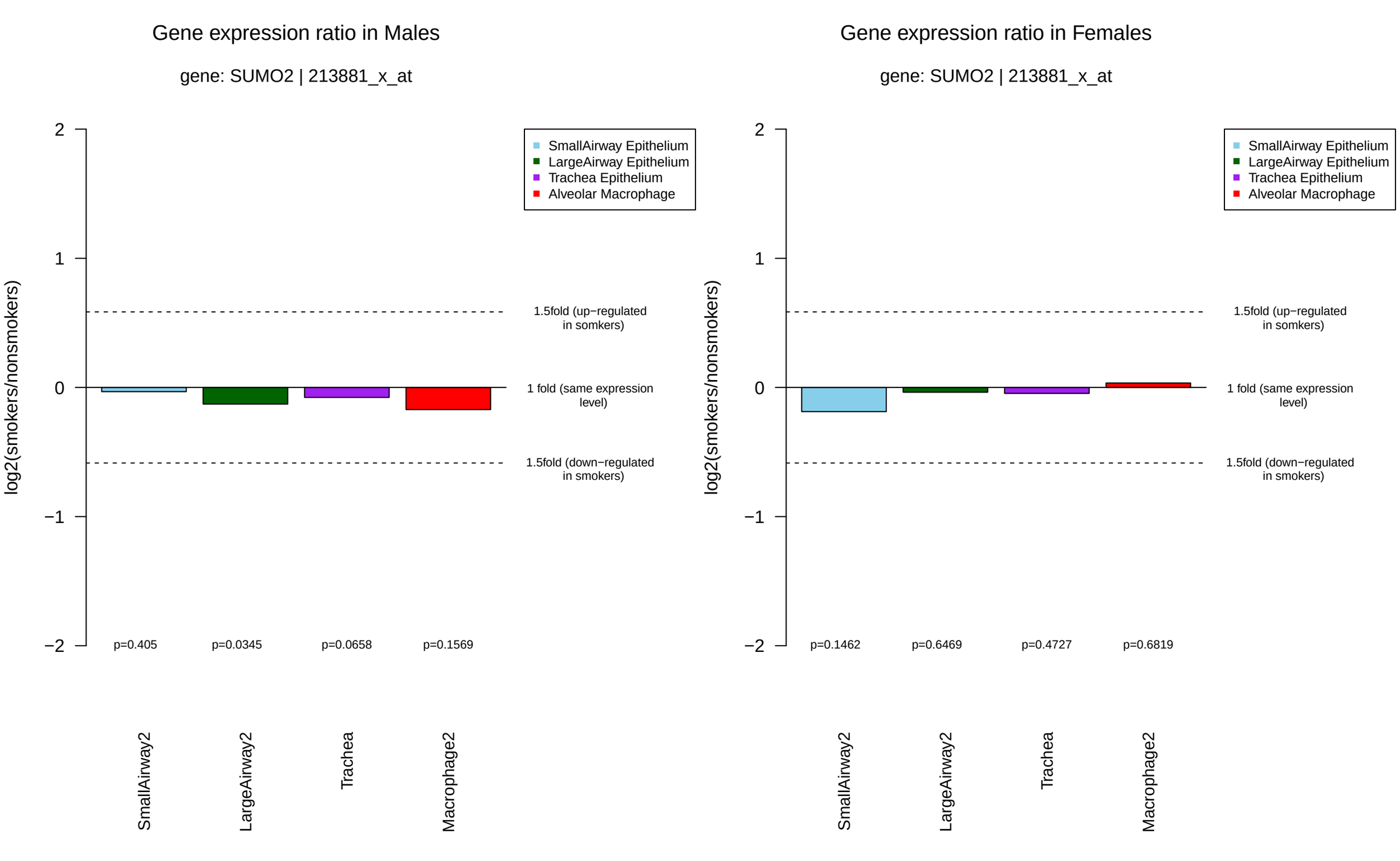
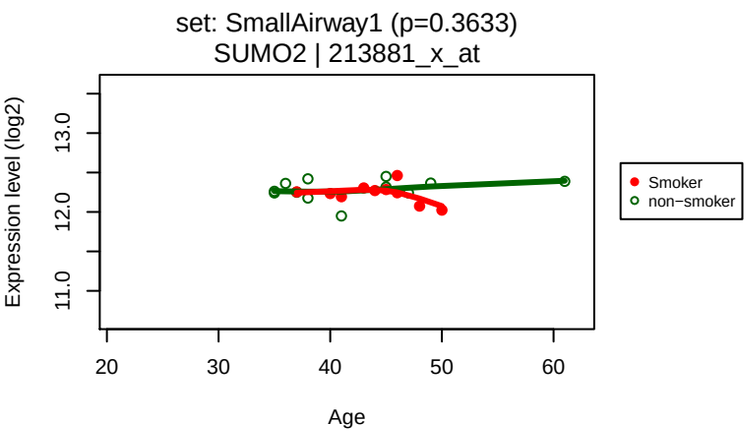
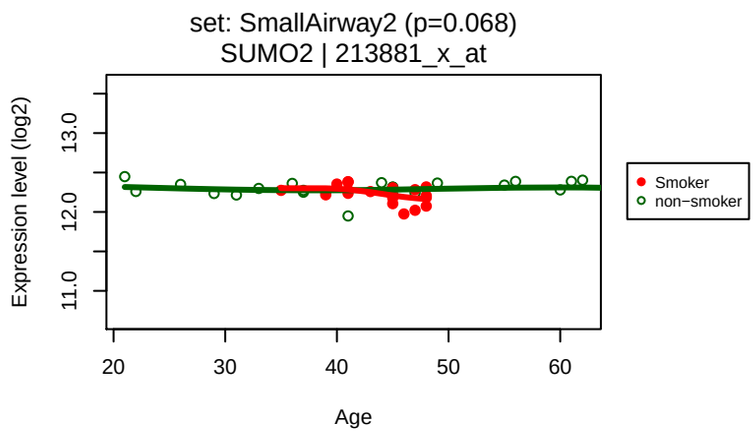
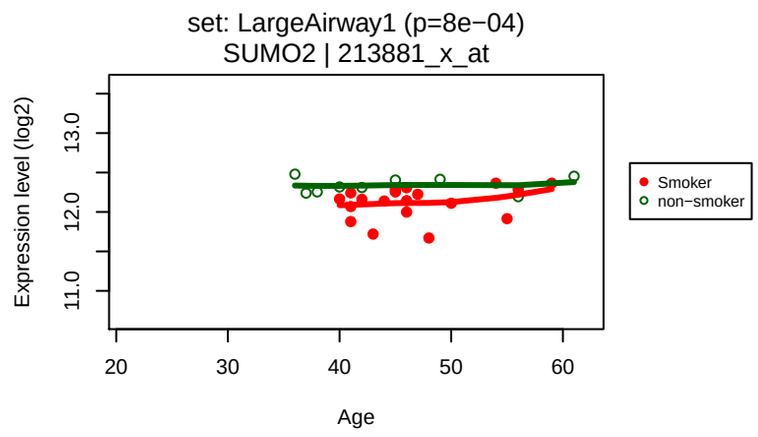

Gene expression of smokers and nonsmokers
gene: SUMO2 | 213881_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3633
p=0.068
p=0.0021
p=8e−04
p=0.0273
p=0.0523
p=0.2939
p=0.2252
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

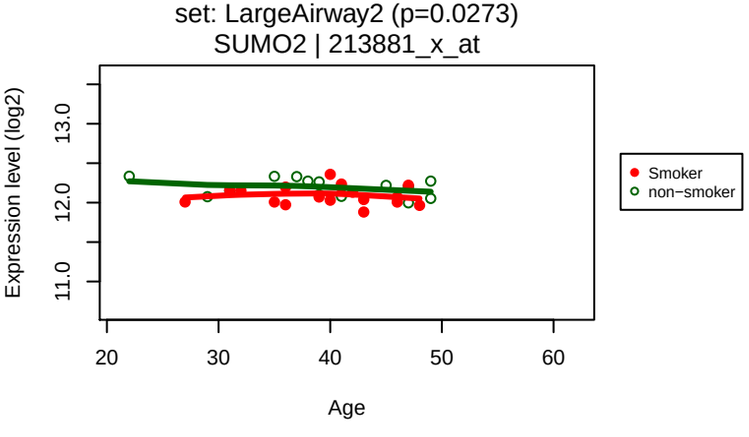

Gene expression ratio (smokers/nonsmokers)
gene: SUMO2 | 213881_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3633
p=0.068
p=0.0021
p=8e−04
p=0.0273
p=0.0523
p=0.2939
p=0.2252
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

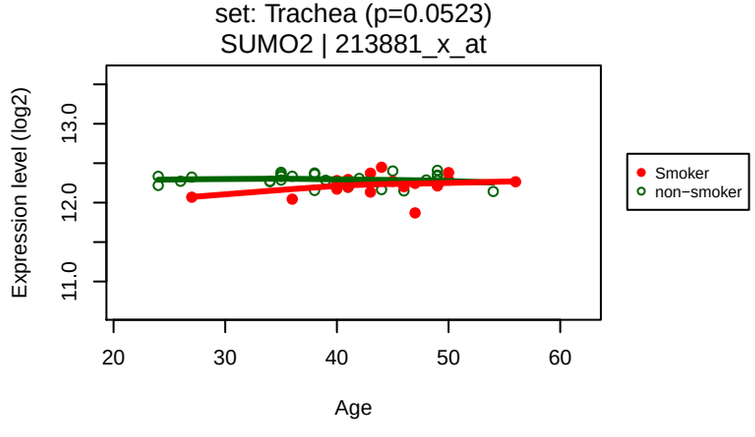

Gene expression ratio in Males
gene: SUMO2 | 213881_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.405
p=0.0345
p=0.0658
p=0.1569
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SUMO2 | 213881_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1462
p=0.6469
p=0.4727
p=0.6819
SmallAirway2
LargeAirway2
Trachea
Macrophage2

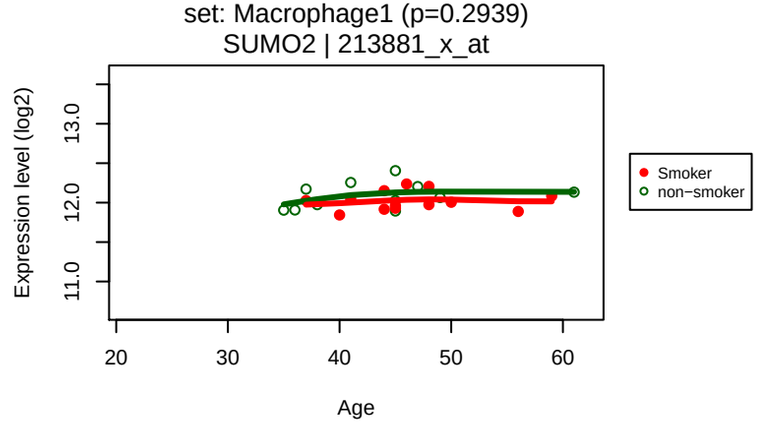

set: SmallAirway1 (p=0.3633)
SUMO2 | 213881_x_at
Expression level (log2)
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker

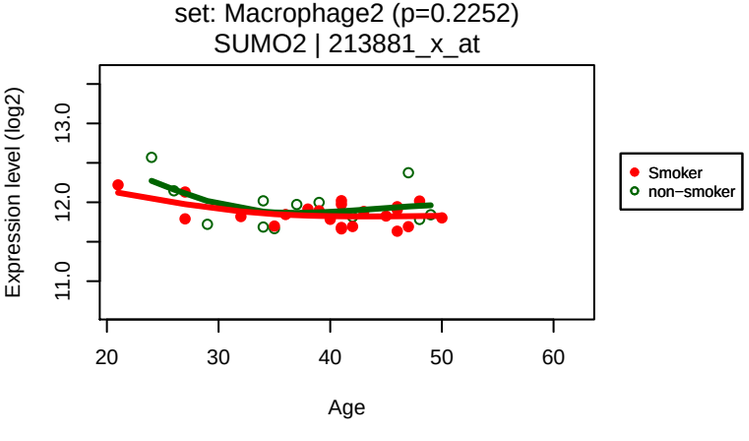

set: SmallAirway2 (p=0.068)
SUMO2 | 213881_x_at
Expression level (log2)
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker

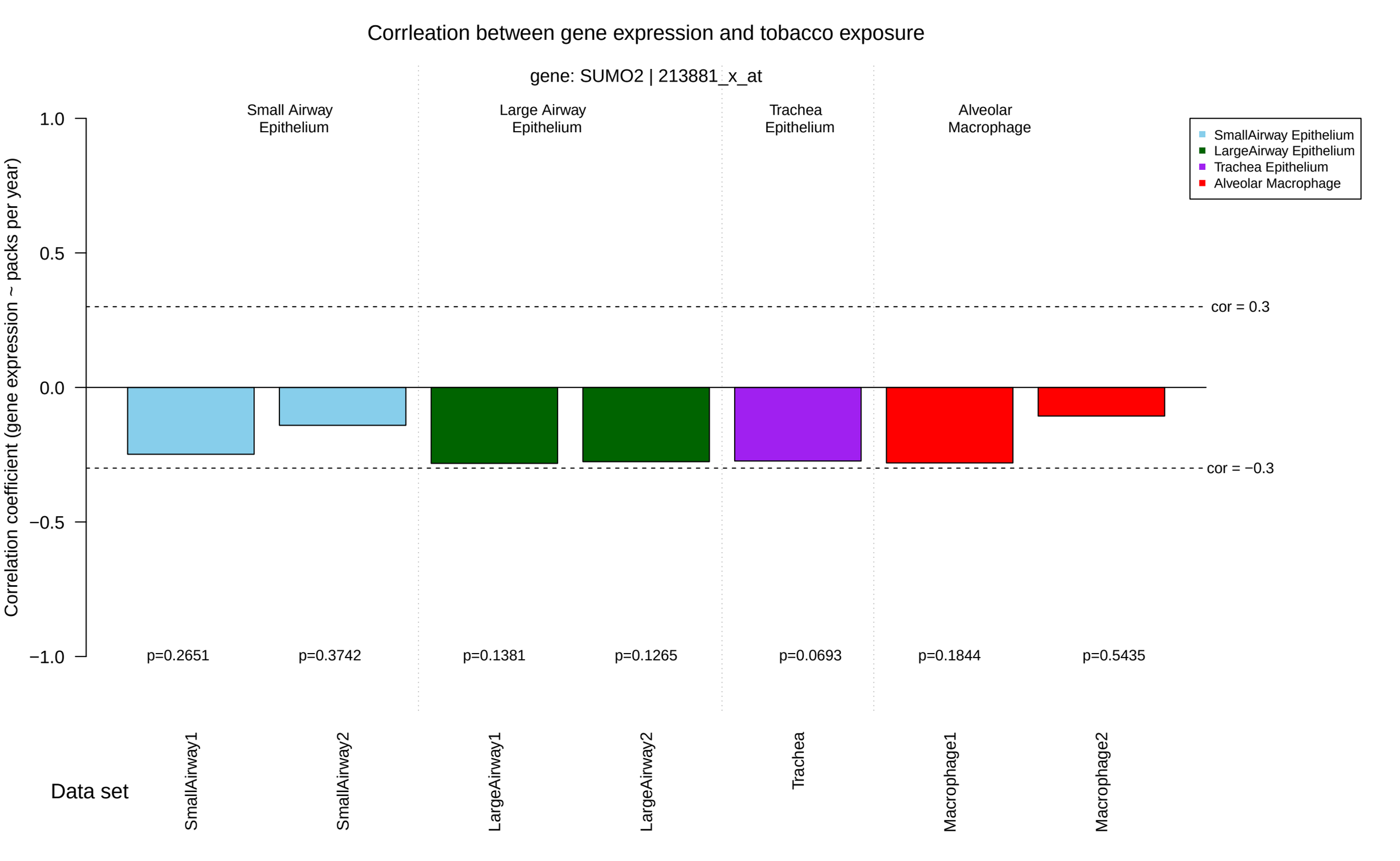

set: LargeAirway1 (p=8e−04)
SUMO2 | 213881_x_at
Expression level (log2)
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0273)
SUMO2 | 213881_x_at
Expression level (log2)
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0523)
SUMO2 | 213881_x_at
Expression level (log2)
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2939)
SUMO2 | 213881_x_at
Expression level (log2)
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2252)
SUMO2 | 213881_x_at
Expression level (log2)
11.0
12.0
13.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SUMO2 | 213881_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2651
p=0.3742
p=0.1381
p=0.1265
p=0.0693
p=0.1844
p=0.5435
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2